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Mobile computing

CIA-1

31/1/24

1) A) cellular network area = 1200 Km2
cell radius = 1.8 Km
Spectrum = 48MHz
reuse cluster sizes N=7, N=4
Reduction = 15%.

(a) calculate number of cells.

$$A_{cell} = \frac{3\sqrt{3}}{2} R^2$$

$$= \frac{3\sqrt{3}}{2} (1.8)^2 = 2.108 (1.8)^2$$

$$A_{cell} = 8.418 \text{ Km}$$

total ~~available channels~~:-
no. of cells

$$\text{Total} = \frac{1200}{8.418} = 142.55$$

(b) total available channels

$$\text{channels} = \frac{48 \times 1000}{200} = 240$$

15% reduction

$$= 240 (1 - 0.15)$$

$$= 204$$

(c) theoritical channels Per cell:-

$n=7$

$$\frac{\text{Total channels}}{N} = \frac{240}{7} = 34.2$$

$n=4$

$$\frac{\text{Total channels}}{N} = \frac{240}{4} = 60$$

effective channels Percell:-

$n=7$

$$\frac{\text{Effective channels}}{N} = \frac{204}{7} = 28.8$$

$n=4$

$$\frac{\text{Effective channels}}{N} = \frac{204}{4} = 51$$

(d) reuse distance

$$D = R\sqrt{3N}$$

$N=7 \Rightarrow D = 1.8\sqrt{3(7)} = 1.8\sqrt{21}$

$$D = 8.25$$

$N=4 \Rightarrow D = 1.8\sqrt{3(4)} = 1.8\sqrt{12}$

$$D = 6.24$$

(e) reuse ration of $Q=4$ is high capacity reuse ration of $Q=5$ is low capacity.

2)

PLMNs . Mobile network = 40 Min

Ambulance . crosses . cell = 15 and boundaries at . 90 Km/h

PLMN handoff requires 120ms

NNS . Path switching . = 80 ms .

Additional authentication = 350 ms .

roaming registration = 250ms .

recovery . = 400ms .

(a) PLMN handoff delay

$T_{normal} = 120 + 80 = 200$ ms

15 handoff

$= 15(200)$.

$= 3000$ ms

$= 3$ s

(b) recovery delay $= 3(400)$

$= 1.2$ s

roaming related delay

$T_{roaming} = 350 + 250 = 600$ ms

2 translators $= 2(600) \Rightarrow 1.2$ s

(c) communication inter :-

$$\text{Total} = 3 + 1.2 + 1.2$$

$$= 5.4 \text{ s}$$

Potential patient data loss at = 512 Kbps

$$\text{loss} = \frac{512(3)}{5.4} \text{ s}$$

$$= 345.6$$

(d) recovery

$$= \frac{400}{40} \Rightarrow \frac{\text{recovery}}{\text{network}}$$

$$= 10 \text{ ms}$$

3)

(i) BSC Manages = 12 ~~Base~~.

BTS with = 3 sector and 4 trasceivers.

each TRX Provides 8 TDMA

time side Per sector = 2

sector server = 1200 Subscribers.

average making 1.2 call Per hour = 90 sec

Traffic channel require s = 16 kbps

BTS connected to the BSC => 2.048 MBPS

(a) analyze base Sfation sector :-

Time slots Per sector :- 4(8) = 32

Traffic channel Per sector = 32-2
= 30

Per BTS :- 30(3) = 90

BSC = 90(12) = 1080

(b) offered traffic Per sector :-

$$A = \frac{\text{calls per hour} \times \text{Call duration in sec}}{3600}$$

$$A = \frac{1200(1.2)(90)}{3600}$$

$$A = 36 \text{ Erlange}$$

(c) channel utilization

$$U = \frac{\text{offered traffic}}{\text{Available channels}} \times 100$$

$$U = \frac{36}{30} \times 100$$

$$\boxed{U = 120\%}$$

(d) Minimum TRXs required Per sector

80% =>

$$A_{max} = 0.8(8T - 2)$$

no. of TRXs per sector:-

$$36 \leq 0.8(8T - 2)$$

$$45 \leq 8T - 2$$

$$47 \leq 8T$$

$$T \geq 5.875$$

$\therefore T = 6$ TRXs Per Sector.

Utilization:- $\frac{36}{46} \times 100 \Rightarrow 78\%$

3) (ii) Total one way spectrum = 10 MHz

FDMA channel bandwidth = 30 KHz, N = 7

TDMA carrier bandwith = 200 KHz

CDMA " " = 1.25 MHz

CDMA chip rate = 1.2288 MCPS.

Voice data rate = 9.6 kbps.

(a) FDMA capacity =

$$C_{FDMA} = \frac{10000}{30} = 333.33$$

Channel Per cell =

$$C_{cell} = \frac{333.33}{7} = 47.62$$

Total user Per cell =

47.62.

(b) TDMA capacity =

$$C_{TDMA} = \frac{10000}{200} = 50.$$

Channel Per cell =

$$C_{cell} = \frac{50}{5} = 10,$$

(c) CDMA Capacity:-

$$C_{CDMA} = \frac{10000}{1.25} = 7500$$

Channel Per Cell =

$$C_{cell} = \frac{7500}{100} = 75$$

(d) CDMA users per carrier:-

$$K = \frac{G_P}{(E_b/N_0)}$$

$$K = \frac{128(0.6)}{8(0.4)}$$

Per cell = $8(24) = 192$

192 user Per cell

(e) Capacity comparision:-

Technique.	User Per cell
FDMA.	47.62
TDMA	100
CDMA	75

4)

Health care network transfer = 240

size = 18 MB

Theoratical data rate = 20 Kbps

reduces capacity = 18%.

congestion limits effective is = 85% and 12%

losting time = 25 sec.

LMTS transfer time is = 384 Kbps.

(a) comparing transerfer Percentage =

data rate% = $\frac{240}{20}$ = 12 Kbps

size = 12 × 18

= 216 MB

(b) evaluate whether GPRS need the 12-hours emergency.

GPRS = $\frac{\text{LMTS}}{\text{losting time}} \times 100$

$= \frac{384}{25} \times 100^{4}$

$384 \times 4 = 1536$

GPRS = 1536 Kbps

(c) Network Performance

$$NP = \frac{\text{transfer time}}{\text{data rate}} = \frac{384}{20} = 19.2 \text{ KbPs}$$

NP = 19.2 KbPS.

(d) Multiple access techniques and health care application.

* There is an loots of applications are in current generation.
* Mainly Network cells to be activate first to call an ambulance in emergency time.
* Bank transactions also crucial in application to pay medicare or some other things
* Network cells mainly active to run and open the application.

(5) Numerical solution:-

Data per device = 120 GB x 1000 . MB = 120000MB

120000/2000 . = 60 MB

= 480 MB

(b) GSM transwer time

$$\frac{480\ MB \times 1000}{9.6} = 50000\ s$$

= 13.89 hours

(c) GPRS transwer

= 480mb x 1.10 x 10000 / 64 kbps

= 8250 s

= 2.29 hours